# Logan H. Jones

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## Research Interests

Galaxy evolution & formation; UV through mm spectroscopy; molecular gas tracers; metallicity evolution of galaxies, incl. stellar abundances; obscured star formation; cosmic reionization and Lyman continuum escape

## Technical Skills

**Languages:** Python, SQL
**Packages:** *astropy*, *numpy*, *scipy*, *emcee*, *pytest*, *multiprocessing*
**Astronomical Tools & Systems:** GILDAS; *jwst*; CalCOS; CASA; IRAF
**General Tools & Systems:** Git, GitHub CI workflows, bash, Sphinx, LaTeX, MS Office, Tableau
**Techniques and Other Skills:** Spectral energy distribution fitting; image processing; spectral data processing; uncertainty propagation; Markov-chain Monte Carlo methods; data visualization; model/pipeline development; storytelling in science communication
**Successful proposals as PI (up to Summer 2026):** ALMA (2023.1.01671.S); NOEMA, 4 proposals. WIYN, 4 proposals. SALT/RSS, 1 proposal.
**Successful proposals as co-I (up to Summer 2026):** JWST, 1 GO proposal (PID 4278). HST, 4 GO proposals (PID 17490, 17515, 17180, 17821) and 1 mid-cycle proposal (PID 16504). Swift, 1 proposal. VLA, 2 proposals. ALMA, 1 proposal. IRAM & NOEMA, 3 proposals.
**Observing:** Classical mode observer at KPNO (in-person and remote) and at Keck (remote). Semiclassical mode observer at JCMT.

## Professional Experience

**Space Telescope Science Institute**
*Postdoctoral researcher* *Oct. 2024 – Sept. 2026*

- Pixel-level cleaning & analysis of JWST imaging to understand extragalactic PAH emission as a function of environment in M101. Homogenized auxiliary datasets of varying quality to complement analyses.
- Created and tested novel continuum modeling methods in Python, with broad applicability for similar (PHANGS-like) datasets.
- Obtained JWST-resolution-matched NOEMA CO observations (PID S25BL, Co-PI L. Jones) of HII regions in M101.
- Awarded 400k core-hours of national supercomputing resources via NSF ACCESS in support of the above work.

*Postdoctoral researcher* *Sept. 2021 – Oct. 2024*

- Developed the Python package `SESAMME`, incl. tutorial notebooks and documentation, for measuring integrated properties of star clusters using MCMC methods.
- Implemented the Togi & Smith (2016) $H_2$ excitation model as the Python package `H2Powerlaw`, incl. tutorial notebooks and documentation, to accelerate scientific discovery with JWST.
- Co-spatial comparisons of young star clusters, atomic gas (ALMA program 2023.1.01671.S, PI L. Jones), and warm $H_2$ from JWST/MIRI spectroscopy.

- Co-spatial comparisons of CO and WR-bearing star clusters to understand N/O vs O/H abundances (NOEMA PID W22CC, Co-PI L. Jones)

**University of Wisconsin – Madison**
*Graduate research assistant* *Sept. 2016 – Aug. 2021*
- Planned semesterly observing campaigns (optical spectroscopy, Keck; single-dish submm imaging, JCMT) as the project steward for long-term research programs.
- Identified rare or unusual astrophysical targets using archival and proprietary data and a combination of automated and manual filtering.

*All positions*: Reported scientific findings in refereed publications (specialized audience), presentations (academic audience), and public talks (general audience); documented code (in-line and public ReadTheDocs manuals).

## Education

**University of Wisconsin-Madison** *M.S. & Ph.D., Astrophysics | Aug. 2021*
GPA: 3.7 | Thesis: From Dusty to Dust-free: Panchromatic Extreme Star Formation in Intermediate-to High-Redshift Galaxies
**University of Arkansas** *B.S., Physics, summa cum laude | May 2016*
GPA: 4.0 | Thesis: Extending the SMBH Mass-Pitch Angle Relation to Moderate Redshifts

## Invited Talks and Conference Contributions

Five most recent invited/contributed talks and/or conference posters shown below.

Exploring the Aromatic Universe *Poster, July 2026*
STScI Spring Symposium: The Metal-poor Frontier *Poster, May 2026*
Dusty Universe 2025 *Poster, Nov. 2025*
New Data that Challenge Underlying Assumptions in Early Galaxy Evol. *Contr. talk, Aug. 2025*
STScI Spring Symposium: Inter+Stellar *Poster, May 2025*

## Service & Engagement

ALMA2040 Working Groups *May 2026 – present*
STScI Research Computing Forum *Mar. 2026 – Sept. 2026*
STScI Science Staff Retreat, postdoc rep. *Sept. 2022, '23, Feb. 2026*
STScI Spring Symposium Local Org. Cmte. *Apr. 2025 & 2026*
Panel Support Scientist, JWST Cycles 2 – 5 Time Allocation Cmte. *Apr. 2023, Jan. 2024, '25, '26*
STScI Spring Symposium Science Org. Cmte. *Apr. 2024*
Mentorship & Peer Mentoring *Various, 2017 – 2023*
— Mentored three undergraduate students & peer-mentored junior graduate student as Ph.D. candidate. Co-mentorship of summer research intern as postdoc.
Panel Leveler, Hubble Space Telescope Cycle 30 Time Allocation Cmte. *June 2022*
Skype a Scientist Presenter *1 – 3/semester, 2020 - present*
UW-Madison teaching assistant *Fall 2020*
— ~350 student, all-virtual introductory course, including the development of project-based group assessments.

UW Astronomy Graduate Lectures for Undergraduates *Fall 2017 & 2018*
WI Institute for Discovery volunteer *Various, 2017 – 2021*
— Event worker for WI Sci. Festival and Jr. Science Cafe programs; Booth coordinator for two astronomy events through the WID Saturday Science program.

# BIBLIOGRAPHY

**Refereed Publications**

- JWST/MIRI-MRS view of the metal-poor galaxy CGCG 007-025: the spatial location of PAHs and very highly ionized gas

Garcia Del Valle-Espinosa, M., Mingozzi, M., ..., **Jones, L. H.**, et al. 2026, ApJL, 998, 45

- A JWST/MIRI View of the ISM in M83: I. Resolved Molecular Hydrogen Properties, Star Formation, and Feedback

**Jones, L. H.**, Hernandez, S., Smith, L. J., Togi, A., et al. 2025, ApJ, 987, 142

- Exploring the mysterious high-ionization source powering [Ne V] in high-z analog SBS0335-052E with JWST/MIRI

Mingozzi, M., Garcia Del Valle-Espinosa, M., ..., **Jones, L. H.**, et al. 2025, ApJ, 985, 253

- JWST/MIRI Detection of [Ne V] and [Ne VI] in M83: Evidence for the Long Sought-after Active Galactic Nucleus?

Hernandez, S., Smith, L. J., **Jones, L. H.** 2025, ApJ, 983, 154

- Spectroscopic Confirmation of a Massive Protocluster with Two Substructures at z ≈ 3.1

Nicandro Rosenthal, M. J., Barger, A. J., Cowie, L. L., **Jones, L. H.**, et al. 2025, ApJ, 979, 247

- Pushing the limits of the Cosmic Origin Spectrograph (COS) with an optimized background correction

Hernandez, S., Igoshev, A., Taylor, J., Sahnow, D., & **Jones, L. H.** 2024, PASP, 136, 4504

- Simultaneous Estimates of Star-cluster Age, Metallicity, Mass, and Extinction (SESAMME) I. Software First-look and Validation

**Jones, L. H.**, Hernandez, S., Smith, L. J., et al. 2023, ApJ, 959, 20

- Dissecting the Mid-Infrared Heart of M83 with JWST

Hernandez, S., **Jones, L. H.**, Smith, L. J., Togi, A., et al. 2023, ApJ, 928, 124

- A Submillimeter Survey of Faint Galaxies Behind Ten Strong Lensing Clusters

Cowie, L. L., Barger, A. J., Bauer, F., Chen, C.-C., **Jones, L. H.**, et al. 2022, ApJ, 939, 5

- Chemical Abundances of Young Massive Clusters in NGC 1313

Hernandez, S., Winch, A., Larsen, S., James, B. L., & **Jones, L. H.** 2022, AJ, 164, 89

- A Submillimeter Perspective on the GOODS Fields (SUPER GOODS) V. Deep 450 μm Imaging

Barger, A. J., , Cowie, L. L., Blair, A., & **Jones, L. H.** 2022, ApJ, 934, 56

- Subaru/HSC z-Excess Broadband Selection of Extreme Emission Line Galaxies at z < 1

Rosenwasser, B., Taylor, A. J., Barger, A. J., Cowie, L. L., Hu, E., **Jones, L. H.**, et al. 2022, ApJ, 928, 78

- NOEMA Redshift Measurements of the Brightest Submillimeter Galaxies in the GOODS-N

**Jones, L. H.**, Rosenthal, M. J., Barger, A. J., & Cowie, L. L. 2021, ApJ, 916, 46

- Candidate z ∼ 2.5 Lyman Continuum Sources in the GOODS Fields

**Jones, L. H.**, Barger, A. J., & Cowie, L. L. 2021, ApJ, 908, 222

- z ∼ 2.5 − 3 Ionizers in the GOODS-N Field

**Jones, L. H.**, Barger, A. J., Cowie, L. L., et al. 2018, ApJ, 862, 142

**Non-Refereed Publications and In-progress Works**

- One-Stop Shopping: Pan-Metallicity PAH Benchmarking in M101 I. Data Overview and Continuum Subtraction Methodology

**Jones, L. H.**, Clark, C. J. R., et al. In prep.

- Spatially Resolving the Fundamental Elements of Reionization in Galaxies

Xu, X., McCandliss, S., ... **Jones, L. H.**, et al. arXiv eprint 2507.01312

- Simultaneous Estimates of Star-cluster Age, Metallicity, Mass, and Extinction (SESAMME) II. Stellar and Gas-phase Metallicities in Diverse Galaxies

**Jones, L. H.**, Hernandez, S., Smith, L. J., et al. In prep.